

Computer Programming Lab (EE-170L)

Laboratory Report

Lab Title: Variables and Basic C++ Programs

Course: Computer Programming Lab (EE-170L)

Instructor: Rizwan Sir

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Student Name: Hidayat Ullah

Date: _____

1. Objective

The objective of this lab is:

- To learn how to declare different types of variables in C++
 - To find memory size, memory location, and values of variables
 - To understand floating-point variables
 - To perform temperature conversion using C++
 - To understand basic syntax and output in C++
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2. Theory

A variable is a container used to store data in memory. Each variable has:

- Name
- Data type
- Value
- Memory location
- Memory size

Common data types in C++:

- int → stores integer values
- float → stores decimal values

- double → stores large decimal values
- char → stores single character
- bool → stores true or false

The sizeof() operator is used to find memory size of a variable.

The address-of operator (&) is used to find memory location.

Temperature conversion formula:

$$F = (C \times 9/5) + 32$$

3. Software / Tools Used

- Visual Studio Code
 - C++ Compiler (MinGW / g++)
 - Windows Operating System
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4. Procedure

1. Open Visual Studio Code
 2. Create new file with extension .cpp
 3. Write the C++ program
 4. Compile the program
 5. Run the program
 6. Observe the output
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5. Task 1: Declare variables and find size, location and values

C++ Code

```
#include <iostream>
using namespace std;
```

```
int main()
```

```
{
    int a = 10;
    float b = 5.5;
    double c = 20.99;
    char d = 'A';
    bool e = true;

    cout << "Integer value: " << a << endl;
    cout << "Size of int: " << sizeof(a) << " bytes" << endl;
    cout << "Location of int: " << &a << endl;

    cout << endl;

    cout << "Float value: " << b << endl;
    cout << "Size of float: " << sizeof(b) << " bytes" << endl;
    cout << "Location of float: " << &b << endl;

    cout << endl;

    cout << "Double value: " << c << endl;
    cout << "Size of double: " << sizeof(c) << " bytes" << endl;
    cout << "Location of double: " << &c << endl;

    cout << endl;

    cout << "Char value: " << d << endl;
    cout << "Size of char: " << sizeof(d) << " bytes" << endl;
    cout << "Location of char: " << (void*)&d << endl;

    cout << endl;

    cout << "Bool value: " << e << endl;
    cout << "Size of bool: " << sizeof(e) << " bytes" << endl;
    cout << "Location of bool: " << &e << endl;

    return 0;
}
```

Output

Integer value: 10

Size of int: 4 bytes

Location of int: 0x61ff08

Float value: 5.5

Size of float: 4 bytes

Location of float: 0x61ff04

Double value: 20.99

Size of double: 8 bytes

Location of double: 0x61fef8

Char value: A

Size of char: 1 bytes

Location of char: 0x61fef7

Bool value: 1

Size of bool: 1 bytes

Location of bool: 0x61fef6

6. Task 2: Floating point variable pi

C++ Code

```
#include <iostream>
using namespace std;

int main()
{
    float pi = 3.14;

    cout << "Value of pi is: " << pi << endl;

    return 0;
}
```

Output

Value of pi is: 3.14

7. Task 3: Celsius to Fahrenheit Conversion

C++ Code

```
#include <iostream>
using namespace std;

int main()
{
    float celsius, fahrenheit;

    cout << "Enter temperature in Celsius: ";
    cin >> celsius;

    fahrenheit = (celsius * 9/5) + 32;

    cout << "Temperature in Fahrenheit is: " << fahrenheit << endl;

    return 0;
}
```

Output Example

Enter temperature in Celsius: 25
Temperature in Fahrenheit is: 77

8. Explanation of Code

- `#include <iostream>` → used for input and output
- `using namespace std` → allows use of `cout` and `cin`
- `int main()` → main function where program starts

- Variables declared using different data types
 - sizeof() → finds memory size
 - & operator → finds memory location
 - cin → takes input from user
 - cout → displays output
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9. Result / Conclusion

In this lab, we learned:

- How to declare variables in C++
- How to find memory size and location
- How to use floating-point variables
- How to convert temperature from Celsius to Fahrenheit
- How to write and execute basic C++ programs